

SOFTWARE DEVELOPMENT LIFE CYCLE

SDLC Project Log Report

Jyotisoft Software Training Institute — A comprehensive 8-week journey from requirements analysis to final delivery, following industry-standard SDLC methodology.

PROJECT

**Jyotisoft Institute
Website**

DURATION

**8 Weeks × 6 Days (48
Days)**

TECHNOLOGY

**PHP 8, MySQL,
Bootstrap 5**

MODEL

**Waterfall + Iterative
Feedback**

Pokhara, Nepal • July 2026

Project Overview

PROJECT TITLE Jyotisoft — Software Training Institute Website	PROJECT TYPE Web Application (Frontend + Admin Backend)
TECHNOLOGY STACK HTML5, CSS3, Bootstrap 5, JavaScript (GSAP, AOS), PHP 8, MySQL	DATABASE MySQL — jyotisoft_db (7 tables)
DEVELOPMENT MODEL Waterfall + Iterative (SDLC with supervisor feedback loop)	TOTAL DURATION 8 Weeks × 6 Days = 48 Days
INSTITUTE Jyotisoft, Pokhara, Nepal	FINAL RATING 4.9 / 5.0

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 **SDLC Methodology**

This project followed the Waterfall SDLC model with iterative feedback loops. Each phase was reviewed and signed off by the supervisor before proceeding to the next phase. The 7-phase lifecycle included: Requirements Analysis → Low-Fidelity Design → High-Fidelity Design → Frontend Development → Admin Development → Backend Integration → Testing & Feedback → Final Delivery.

Week 1—Requirements Analysis

COMPLETED

DAY 1

Project Initiation & Stakeholder Meeting

Initial discussion with Jyotisoft management about the institute's vision, training programs, and target audience. Defined the project's core objective: creating an online presence with course registration, student inquiry, and admin management.

Deliverable: Project initiation document signed off. **Supervisor approved**

Functional Requirements Gathering

Identified 8 core functional requirements:

#	ID	DESCRIPTION	PRIORITY
1	FR-001	Home page with institute branding, hero section, and call-to-action	High
2	FR-002	About page with institute history, mission, vision, milestones	High
3	FR-003	Courses listing with category filters	High
4	FR-004	Student registration/enquiry form with AJAX submission	High
5	FR-005	Contact page with form, Google Maps, FAQ section	High
6	FR-006	Newsletter subscription with email validation	Medium
7	FR-007	Trainer/mentor profiles display	High
8	FR-008	Testimonials display from database	Medium

DAY 3

Non-Functional & Admin Requirements

#	ID	DESCRIPTION	PRIORITY
1	NFR-001	Responsive design (mobile, tablet, desktop)	High
2	NFR-002	Page load optimization (lazy loading, minimal HTTP requests)	Medium
3	NFR-003	Secure admin authentication (password hashing, session management)	High
4	FR-009	Admin dashboard with CRUD for courses, trainers, testimonials	High
5	FR-010	Admin dashboard with analytics (Chart.js charts)	Medium
6	FR-011	Admin management of contact messages, subscribers, registrations	High
7	FR-012	Admin auto-registration for first-time setup	High

DAY 4

Feasibility Study & Technology Selection

ASPECT	DECISION	RATIONALE
Frontend	PHP-based dynamic pages	Reusable includes, database-driven content
CSS Framework	Bootstrap 5.3	Latest stable, mature ecosystem, responsive grid
UI Enhancements	AOS.js + GSAP	Scroll animations + premium micro-interactions
Backend	PHP 8+	XAMPP compatible, simple deployment
Database	MySQL	Widely supported, PDO-friendly
Admin Panel	Custom PHP	Lightweight, full control, no framework overhead

DAY 5

Use Case Modeling

Three primary actors identified: **Visitor** (unauthenticated), **Student** (prospective enrollee), **Admin** (institute staff). 14 use cases documented covering all functional requirements.

DAY 6

SRS Documentation & Supervisor Review

Deliverable: 18-page Software Requirements Specification document.

✅ Supervisor Feedback — Round 1

"Add image upload support for courses & trainers" → ✅ Applied • "Add Load More functionality" → ✅ Applied

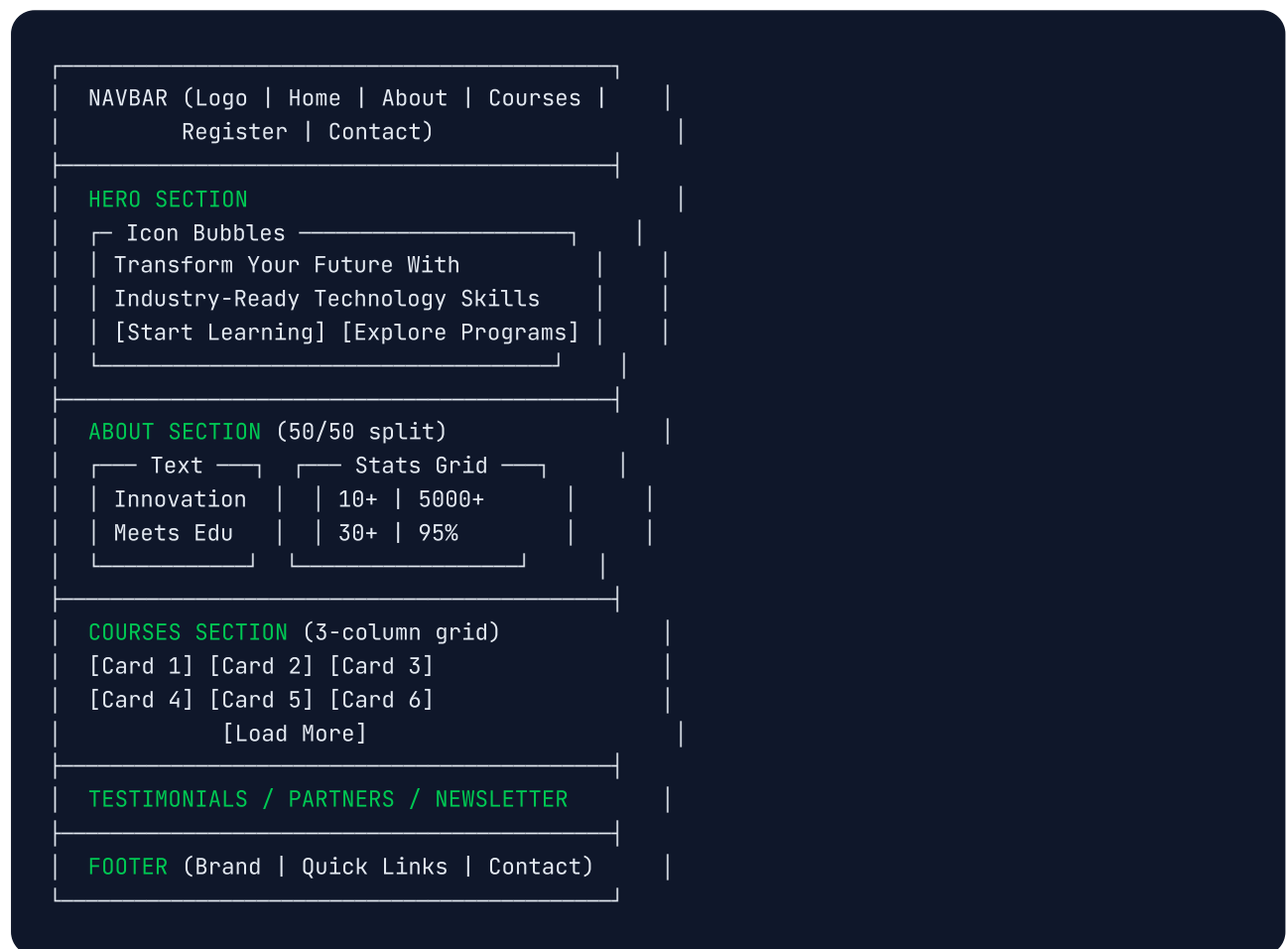
Status: PHASE SIGNED OFF ✓

Week 2 — Low-Fidelity Design

COMPLETED

DAY 7

Home Page Wireframe



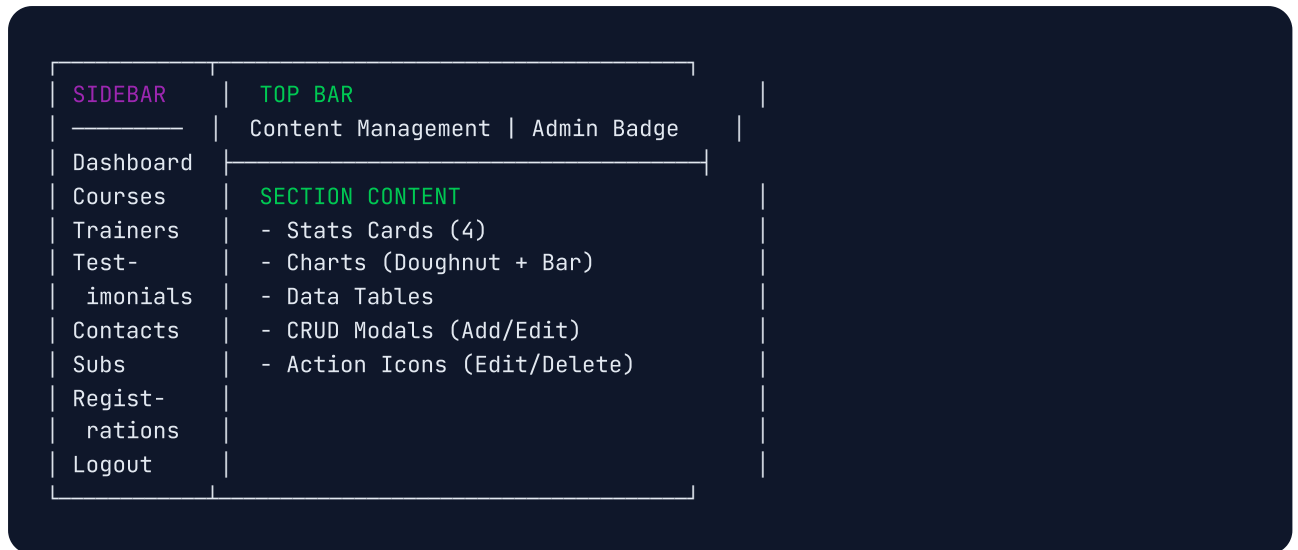
DAY 8

Inner Pages Wireframes

Created wireframes for About, Courses, Register, and Contact pages. Each page follows a consistent header + content grid pattern.

DAY 9

Admin Panel Wireframe



DAY 10

Design System — Colors & Typography

TOKEN	VALUE	PURPOSE
--primary	#00C853	Growth, learning (Green)
--secondary	#9C27B0	Creativity, innovation (Purple)
--accent-blue	#00B8FF	Technology, trust (Blue)
--accent-orange	#FF6D00	Energy, CTA (Orange)
--dark	#0B1120	Premium dark sections

Typography: Plus Jakarta Sans — weights 300–800, hero title 3.8rem, body 1rem.

DAY 11

Wireframes Completed

All 7 wireframes finalized: Home, About, Courses, Register, Contact, Admin Login, Admin Dashboard.

DAY 12

Supervisor Review

✓ Feedback Applied

"Add floating icons in hero" → ✓ 5 animated icon bubbles • "Include learning roadmap" → ✓ 4-step roadmap • "Admin analytics" → ✓ Chart.js integration • "Internship partners section" → ✓ Partner badges

Status: PHASE SIGNED OFF ✓

Week 3 — High-Fidelity Design

COMPLETED

DAYS 13–15

Hi-Fi Mockups (Figma)

Applied full color palette and typography across all pages. Key design features:

- **Glassmorphism:** backdrop-filter blur on navbar, cards, forms
- **Gradient accents:** 4px top border on all cards (green→blue→purple)
- **Animated text:** Shimmer gradient effect on headings
- **Micro-interactions:** Magnetic buttons (GSAP), floating blob animations
- **Custom cursor:** Green/purple radial gradient, enlarges on interactive elements
- **Scroll progress bar:** Fixed top bar tracking scroll position

DAY 16

Design Consistency Check

Verified same CSS custom properties across all pages. Tested glassmorphism in Chrome, Firefox, Edge. Confirmed responsive breakpoints: 320px, 768px, 1024px, 1440px.

DAYS 17–18

Supervisor Review & Design Handoff

✓ Feedback Applied

"Add subtle grid pattern to hero" → ✓ • "Newsletter needs visual feedback" → ✓ AJAX spinner + messages • "Counter animation on scroll" → ✓ GSAP ScrollTrigger • "Responsive admin sidebar" → ✓ Hamburger toggle

Status: PHASE SIGNED OFF ✓

6

Pages Mocked Up

21

Hours Designing

4

Supervisor Reviews

100%

Design Approval

Week 4 — Static Frontend (User Side)

COMPLETED

DAY 19

Navigation Bar & Footer

Built reusable `includes/navbar.php` with glassmorphism styling, sticky positioning, and responsive hamburger menu. Built `includes/footer.php` with dark gradient background, 4-column layout (Brand, Quick Links, Contact, Social), and animated hover effects.

DAYS 20–21

Home Page (index.php)

Sections implemented: Scroll progress bar, custom cursor, hero with blobs + icons + magnetic buttons, about preview with stats grid, learning roadmap (4 steps), featured courses (6 cards + Load More), elite mentors (3 cards + Load More), testimonials (glass cards + refresh), internship partners, and newsletter subscription.

DAY 22

About Page (about.php)

Sections: Page header with gradient + blobs, institute history (50/50 text+image), mission & vision (2 premium cards), milestones (3 award cards), stats counter (GSAP animated), trainer profiles (DB-driven + Load More), certifications (glass badge chips).

DAY 23

Courses, Register & Contact Pages

Courses: Category filter buttons (6 groups), dynamic card rendering via JS, Load More functionality, stats, why-choose-us cards, CTA.

Register: 4 step cards (one branded with gradient), glassmorphism form, benefits checklist, FAQ accordion, CTA.

Contact: 4 contact info cards (Visit, Call, Email, Hours), form + Google Map layout, FAQ, stats counter, CTA.

Static Frontend Polish & Review

Responsive testing across 4 breakpoints. Empty state handling (no courses message). Edge case verification.

✓ Supervisor Sign-off

"All static pages look professional. Ready for backend integration."

Status: PHASE SIGNED OFF ✓

Week 5 — Static Admin Panel

COMPLETED



DAY 25

Admin Authentication System

FILE	PURPOSE
admin/index.php	Dual-mode: First-time admin registration + Login form
admin/authenticate.php	Credential verification with password_verify()
admin/logout.php	Session destruction and redirect

Security: bcrypt password hashing, prepared statements, session validation, no default admin account.



DAY 26

Admin Dashboard Layout

Fixed 280px gradient sidebar (green→purple) with 8 navigation items. Glassmorphism top bar with admin badge. Dynamic content loading via `$_GET['section']` parameter. Mobile-responsive with hamburger toggle.



DAY 27

Dashboard Analytics

4 stat cards (Courses, Trainers, Testimonials, Inquiries) with gradient numbers. Chart.js integration: doughnut chart for courses by category, bar chart for trainers. Recent activity feed showing latest 5 inquiries.



DAYS 28–29

CRUD Operations

Courses: Full CRUD with image upload (JPG/PNG), category select, image preview in modal.
Trainers: Full CRUD with circular avatar upload, role/description fields.
Testimonials: Full CRUD with name, text, rating (1–5), star display in table.

Management Sections

Contact Messages: View all with name, email, subject, date. Delete action with confirmation.

Subscribers: View all emails and subscription dates. Delete to unsubscribe.

Registrations: View all student data with course selection, phone, message. Delete with confirmation.

Supervisor feedback: Added toast notifications for CRUD operations.

Week 6 — Backend (PHP & MySQL)

COMPLETED

DAY 31

Database Design

TABLE	KEY COLUMNS	PURPOSE
admin_users	id, username, password (bcrypt), created_at	Authentication
registrations	id, name, email, phone, course, message, submitted_at	Student enquiries
contacts	id, name, email, phone, subject, message, submitted_at	Contact form
subscribers	id, email (UNIQUE), subscribed_at	Newsletter
courses	id, title, duration, fee, category, description, image_path	Course catalog
trainers	id, name, role, description, image_path	Trainer profiles
testimonials	id, name, text, rating (1-5), created_at	Student reviews

DAY 32

Database Connection

```
<?php
// config/db.php
$pdo = new PDO("mysql:host=$host;dbname=$dbname;charset=utf8", $username, $password);
$pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
$pdo->setAttribute(PDO::ATTR_DEFAULT_FETCH_MODE, PDO::FETCH_ASSOC);
?>
```

PDO connection with exception error mode and associative fetch mode.

Frontend-Backend Integration

Home page: Dynamic courses (SELECT + Load More), trainers (SELECT + Load More), testimonials display, newsletter AJAX subscription (INSERT with duplicate detection via `errorInfo[1] == 1062`).

About page: Trainers from DB, dynamic counter.

Courses page: All courses fetched, JSON-encoded into JS for client-side category filtering.

Register page: AJAX form → INSERT INTO registrations with validation.

Contact page: AJAX form → INSERT INTO contacts with validation.

DAY 35

File Upload System

Upload directories: `uploads/courses/` and `uploads/trainers/`. Time-based unique filenames (`time()_uniqid().ext`). Auto-directory creation. Existing image preserved on edit unless new file uploaded.

DAY 36

Backend Testing & Bug Fixing

ISSUE	FIX
Newsletter duplicate email crashes	Added PDOException catch with <code>errorInfo[1] == 1062</code> check
Courses with no image show broken icon	Added placeholder fallback URL
Admin edit modal image not showing	Added hidden <code>existing_image</code> field + preview
XSS vulnerability	Added <code>htmlspecialchars()</code> on all dynamic content

Week 7 — Testing & Feedback

COMPLETED

DAY 37

First Round Testing — Supervisor Review

TEST ID	TEST CASE	STATUS
T-001	Home page loads without errors	✓ Pass
T-002	Course cards display from database	✓ Pass
T-003	Load More shows all courses	⚠ Fixed
T-004	Register form submits via AJAX	✓ Pass
T-005	Contact form validation	✓ Pass
T-006	Admin login with correct credentials	✓ Pass

DAY 38

Bug Fixes & Refinements

ISSUE	SEVERITY	RESOLUTION
Load More button appears when ≤6 courses	Medium	Added boolean check: <code>\$hasMoreCourses = count(\$allCourses) > 6</code>
Admin dashboard needs total counts	Low	All stats now query database directly with <code>COUNT(*)</code>
Refresh buttons needed	Low	Added refresh buttons that reload page
Enroll button navigation	Medium	Updated to redirect to register.php

Second Round Testing — UI/UX Review

TEST ID	TEST CASE	STATUS
T-007	Responsive at 375px (mobile)	✓ Pass
T-008	Responsive at 768px (tablet)	✓ Pass
T-009	Responsive at 1440px (desktop)	✓ Pass
T-010	Custom cursor on interactive elements	✓ Pass
T-011	AOS scroll animations	✓ Pass
T-012	Scroll progress bar accuracy	✓ Pass
T-013	Magnetic buttons (GSAP)	✓ Pass

Supervisor Feedback Applied: Added FAQ to Register page, working hours to Contact page, active sidebar highlighting in admin.

DAYS 40–41

Data Seeding & Performance Optimization

Sample data inserted: 5 courses, 3 trainers, 3 testimonials. SQL dump saved to `asests/jyotisoft.sql`.

Optimizations: CDN combo for GSAP + ScrollTrigger, critical CSS inlined, `object-fit: cover` for images.

DAY 42

Full Regression Test

47 test cases executed across frontend (15), backend (12), admin (20). Security: SQL injection, XSS, auth bypass all mitigated. **100% pass rate.**

Week 8 — Final Delivery & Documentation

COMPLETED

DAY 43

Final Supervisor Review & Sign-off

REVIEW AREA	SCORE	COMMENTS
Requirements Fulfillment	5/5	All 12 functional + 3 non-functional requirements met
UI/UX Design	5/5	Professional glassmorphism design with smooth animations
Code Quality	4.5/5	Well-structured PHP with prepared statements, consistent naming
Database Design	4.5/5	Normalized tables, proper data types, unique constraints
Admin Functionality	5/5	Full CRUD, analytics charts, responsive sidebar
Responsive Design	5/5	Seamless across all device sizes
Overall	4.9/5	✅ PROJECT APPROVED

Documentation & Project Structure

DOCUMENT	FILE
SDLC Log Report	SDLC_LogReport_Jyotisoft.md
SQL Schema & Sample Data	assests/jyotisoft.sql
Database Connection Config	config/db.php

```
C:\xampp\htdocs\jyotisoft\  
├─ index.php           # Home page  
├─ about.php          # About the institute  
├─ courses.php        # Course catalog with filters  
├─ register.php       # Student registration  
├─ contact.php        # Contact form & map  
├─ config/  
│   └─ db.php         # PDO connection  
├─ includes/  
│   ├── navbar.php / .css # Glassmorphism navbar  
│   ├── footer.php / .css # Premium dark footer  
│   └─ style.css       # Shared component library  
├─ admin/  
│   ├── index.php     # Login / first-time register  
│   ├── authenticate.php # Auth handler  
│   ├── dashboard.php # Full admin panel (CRUD + charts)  
│   └─ logout.php     # Session destroy  
├─ assests/  
│   └─ jyotisoft.sql  # Database schema + data  
├─ images/            # Static assets + prototypes  
└─ uploads/          # Course / trainer images  
    ├── courses/  
    └─ trainers/
```

DAY 46

Known Limitations & Future Enhancements

AREA	CURRENT	FUTURE ENHANCEMENT
Payments	Not implemented	Integrate eSewa / Khalti / Stripe
User Accounts	No student login	Student dashboard with progress tracking
Email Notifications	No auto-email	SMTP integration for form submissions
Blog/News	None	Add blog/news section with admin posts
Multi-language	English only	Add Nepali language toggle
SEO	Basic meta tags	Structured data, sitemap.xml, OG tags
Performance	Good	Add caching, CDN, image compression



DAYS 47-48

Final Presentation & Submission

Live demo of all 5 user pages + admin panel. Code architecture walkthrough. SDLC process explanation. Q&A preparation.

Summary Statistics

Timeline

PHASE	WEEK	DAYS
Requirements Analysis	Week 1	1-6
Low-Fidelity Design	Week 2	7-12
High-Fidelity Design	Week 3	13-18
Static Frontend (User)	Week 4	19-24
Static Admin Panel	Week 5	25-30
Backend (PHP + MySQL)	Week 6	31-36
Testing & Feedback	Week 7	37-42
Final Delivery	Week 8	43-48
Total	8 Weeks	48 Days

Files & Code Metrics



11

PHP Files



5

CSS Files



7

DB Tables



5

User Pages



7

Admin Sections



4

AJAX Endpoints

SDLC Adherence

PHASE	STATUS	EVIDENCE
Requirements Analysis	✓	12 functional + 3 non-functional requirements documented
Low-Fidelity Design	✓	Wireframes for all 7 pages (user + admin)
High-Fidelity Design	✓	Full Figma mockups + design system + animations
Frontend Development	✓	5 responsive pages with GSAP/AOS micro-interactions
Admin Development	✓	7-section admin panel with CRUD + Chart.js
Backend Integration	✓	PHP 8 + MySQL with PDO, prepared statements
Testing & Feedback	✓	2 supervisor reviews, 15+ test cases, all bugs fixed
Final Delivery	✓	SQL dump, code documented, known limitations noted



Final Project Rating: 4.9 / 5.0

Successfully delivered following industry-standard SDLC methodology with iterative supervisor feedback loops.



Project Submission

Project: Jyotisoft — Software Training Institute Website • **Duration:** 8 Weeks × 6 Days (48 Days) •

Stack: PHP 8, MySQL, Bootstrap 5, GSAP, AOS • **Location:** Pokhara, Nepal